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Development of a compact torus injector system for central fuelling technology of EAST

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Abstract: In this paper, a compact torus (CT) injector system is designed for experimental advanced superconducting tokamak (EAST). The injector system consists of a coaxial electrode and four pulsed power systems and generates self-organized CT with high density and velocity. In addition, the performance of EAST-CTI is evaluated using a series of diagnostics systems. The experimental results show that the velocity, electron density, and number of particles of the CT are 56.0 km/s, 8.73×10^{20} m^{-3}, and 2.4×10^{18}, respectively. The detailed description and experimental results are presented in this paper.

Keywords: Pulsed power; Plasma diagnostics - charged-particle spectroscopy; Plasma diagnostics - probes

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Contents

1 Introduction

With the development of controlled nuclear fusion, the Chinese Fusion Engineering Experimental Reactor (CFETR) proposes two main goals for next-generation fusion devices: achieving a fusion power of 50–200 MW and self-sufficient tritium breeding [1]. Central fueling can considerably increase the fusion power of the core plasma and reduce the demand for tritium. Compared to fueling techniques such as gas puffing [2], pellet injection [3], and supersonic molecular beam injection [4], compact torus (CT) injection is a promising technology for central fueling wherein the CT, which is a carrier of kinetic, thermal, and electromagnetic energy, is formed and accelerated in a coaxial plasma gun. CT is extensively employed in fusion devices for central fueling, such as TdeV [5], STOR-M [6], JFT-2M [7], C-2 [8], and KTX [9]. In addition, the CT injector is used for studying magnetic reconnection [10], magnetic target fusion [11], and space plasma propulsion [12].

To date, few experimental results have been presented for CT injection on medium or large tokamaks. The toroidal field strength of experimental advanced superconducting tokamak (EAST) can reach up to 3.5 T, which is substantially higher than that of other fusion devices [13]; in addition, the distance from the boundary to the center of the plasma is large, which means that CT requires a tighter structure and speed. Therefore, a bench test is required for the injector. In this paper, the EAST-CTI is described in detail, and a series of diagnostics is established by employing the injector system. Furthermore, the performance of the device is evaluated using the data obtained from the measurements.

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2 Description of the injector and diagnostics

2.1 Apparatus

2.1.1 Principles and instruments

The classical structure of the CT injector system includes coaxial cylindrical electrodes, a power supply system, a fast gas input system, and vacuum and acquisition systems, as illustrated in figure 1(a). The bias solenoid provides a bias poloidal field B_{bias}, which links the inner and outer electrodes in the formation region. A neutral gas (helium) is fed into the formation region through eight fast gas valves. Then, a high formation-bank voltage V_f is applied, which breaks down the neutral gas and ionizes it between the coaxial electrodes. This results in a radial current J_r through the ionized plasma, which induces an azimuthal magnetic field B_θ. The plasma cluster is moved in the axial direction by Lorenz force $J_r \times B_\theta$. The bias magnetic field lines pinches off the plasma cluster, forming the poloidal magnetic field of the CT. Subsequently (after several microseconds), the CT reaches the force-free state and yields a long-lived CT configuration, a state known as the "Taylor state" [14]. Then, a high acceleration-bank voltage V_a is applied, and the CT is accelerated by Lorenze force $J_r \times B_\theta$. After being compressed in the compressor, the CT reaches the acceleration region and is continuously accelerated. Finally, CT is punched out from the end of the coaxial electrode.

The schematics and photographs of the EAST-CTI are shown in figure 1. The EAST-CTI consists of a formation region, a compressor, an acceleration region, a gas valve, insulators, and a solenoid. The power supplies of the formation and acceleration regions are connected to the injector via an external flange. The inner and outer electrodes are separated by insulators. The outer shell of EAST-CTI is the only grounding point for all bank systems.

The diameter of the inner and outer electrodes in the formation region is 209 mm and 250 mm, respectively, and the length of both electrodes is 420 mm; the electrodes are constructed using 304 stainless steel. The compressor is 278 mm long, and its radius varies from 50 mm to 125 mm, with a compression ratio of 2.5. The compression process is considered to be adiabatic; therefore, CT with higher density, temperature, and magnetic field can be realized. The diameter of the inner and outer electrodes in the acceleration region are 100 mm and 35 mm, respectively, and the length of both electrodes is 500 mm. The electrodes are constructed using oxygen-free copper, which has high electrical conductivity.

To realize a CT with a spheromak-like plasma ring structure, a solenoid is set inside the inner electrode of the formation region, producing a bias magnetic flux. The solenoid is made of copper wire, which is wound on a polytetrafluoroethylene (PTFE) backbone. The length, thickness, number of turns, inductance, and resistance of the solenoid are 162 mm, 7.6 mm, 810 turn, 80.6 mH, and 6.7 Ω, respectively. To withstand high voltages, the surface of the solenoid is covered with polyimide. The solenoid must maintain a magnetic field for an adequate duration. Typically, the penetration time of the CT does not exceed tens of microseconds, and the magnetic field applied to the CT during this period must be approximated as a quasi-steady field.

Eight gas valves are positioned 45° apart azimuthally to ensure a uniform distribution of pure helium in the formation region; four valves were used in the initial tests. The valve was originally designed by University of Science and Technology of China (USTC), which yields a fast reaction time. The gas valve consists of round pie coils, an aluminum piston, and a spring. One end of the valve is connected to the gas cylinder and the other end is connected to the air intake. When a pulse current is applied to the coil inside the valve, resulting a repulsive force between the coil and piston. The force lifts the piston from the coil allowing the gas to flow into the formation region. The action of the piston can be adjusted by changing the compression of the spring, thus regulating the amount of air intake.

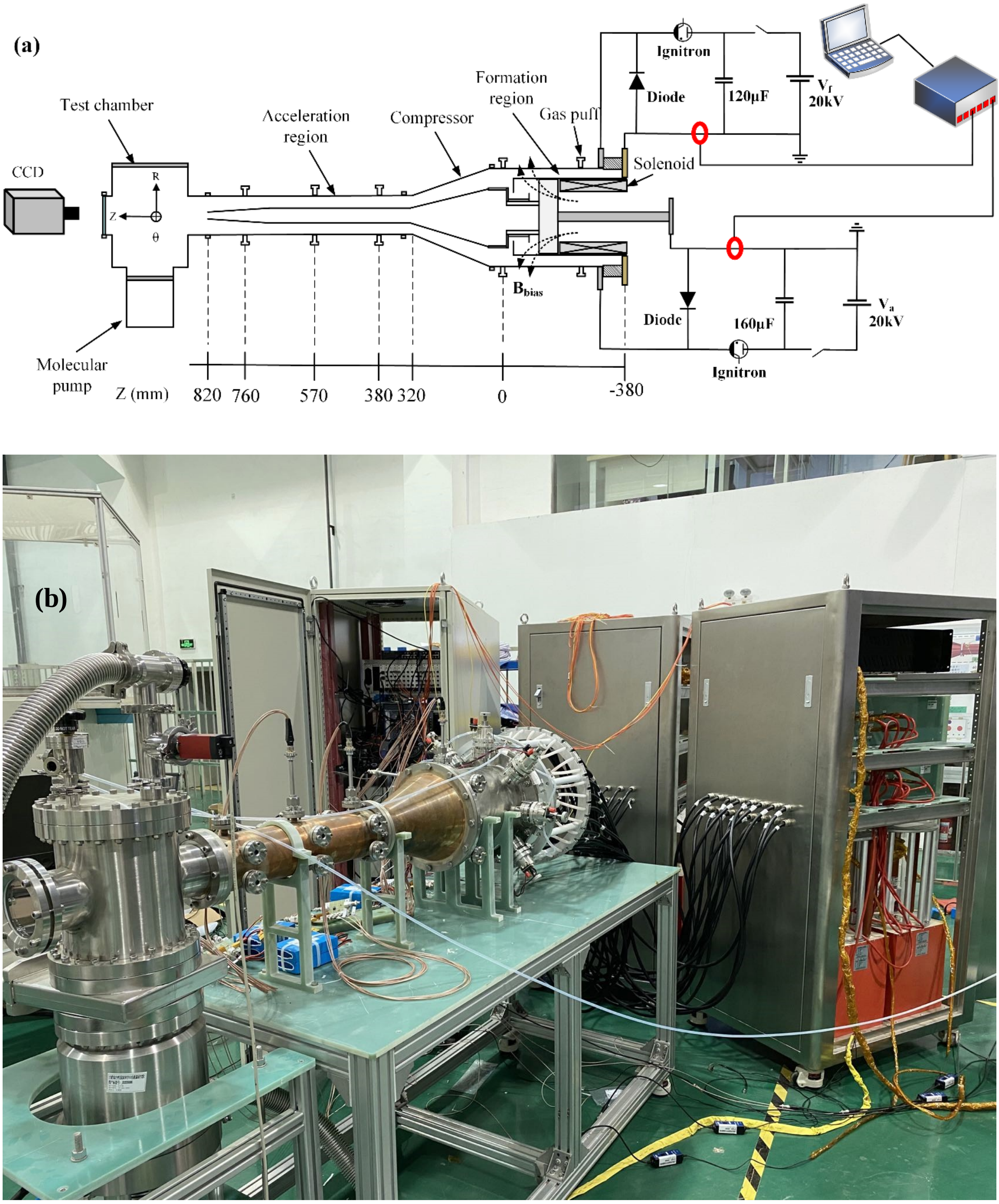

Figure 1. (a) Schematic of the EAST-CTI system, (b) photograph of the EAST-CTI system.

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2.1.2 Power supplies

Table 1. Power supplies of EAST-CTI.

Position	Capacitances (μF)	Maximum voltage (kV)
Formation	120	20
Acceleration	160	20
Bias solenoid	500	2
Gas valve	250	2

The power supply system of EAST-CTI is composed of the formation region, acceleration region, bias solenoid, and gas valve. The capacitances of the formation and acceleration region are 120 μF (four 30 μF capacitors in parallel) and 160 μF (four 40 μF capacitors in parallel), respectively (table 1). The maximum discharge voltage for the main circuits of the formation region and acceleration region is 20 kV, and the maximum energy storage is 24 and 32 kJ, respectively. The formation and acceleration regions are connected to the injector by 16 parallel cables, which considerably reduce the overall inductance of the circuit. The outer shell of EAST-CTI is the only grounding point for all bank systems, and the injector adopts negative high-voltage pulse discharge, which yields superior performance of the CT [15]. A 250 μF capacitor bank is used to drive these gas valves, and the solenoid is drived using a 500 μF capacitor bank. The maximum voltage for both main circuits is 2 kV.

The bias magnetic flux is provided by the solenoid installed in the formation region; the duration of the solenoid magnetic field must be sufficient to allow the magnetic field to penetrate through the electrode. The frequency of bias current f_{bias} required to allow the magnetic field to penetrate through the electrode can be estimated using the equation, $f = \rho/\pi\mu\delta^2$ [16], where ρ, δ, and μ are the resistivity of the conductor, skin depth, and magnetic permeability, respectively. The thickness of the formation electrodes is 3 mm; thus, the frequency f_{bias} must be considerably lower than 20 kHz. The frequency of bias current f_{bias} in the solenoid can be estimated using the equation, $f = 1/2\pi\sqrt{LC}$, where L and C are inductance and capacitance, respectively. The frequency of bias current f_{bias} is approximately 7.9 Hz. Thus, the frequency of the solenoid allows the magnetic field to penetrate the electrodes.

In figure 2(a), at 2 kV discharge voltage, the solenoid current is 137 A. The full width at half maximum (FWHM) discharge waveform of the solenoid is approximately 14 ms, which is much longer than the main circuit discharge time. Simulation using COMSOL yielded the poloidal magnetic flux as 10 mWb and the poloidal magnetic field at the center of the solenoid coil as approximately 0.5 T. The discharge voltage and current waveforms are shown in figure 2(b); at 2 kV discharge voltage, the current is 9000 A and the FWHM of the current waveform is approximately 90 μs, thus meeting the requirements for particle injection and fast response.

2.1.3 Data acquisition and vacuum system

The data acquisition system consists of six high-speed oscilloscope cards PXIe-69834, and the maximum sampling rate is 80 MHz at 16-bit resolution. Each of the six cards has four channels; thus, the system has twenty-four high-speed channels. The six oscilloscope cards are mounted on the oscilloscope chassis PXIe-62590 and connected to the computer via a controller PXIe-63977w.

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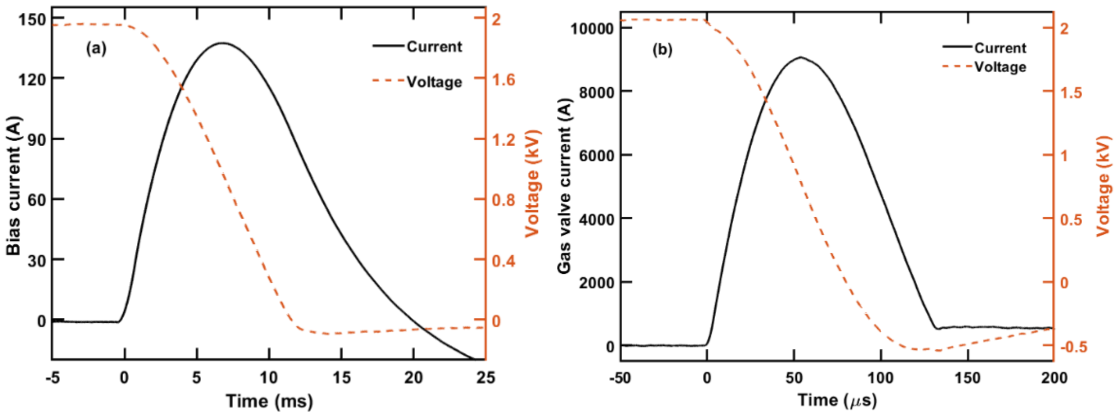

Figure 2. Voltage and current curve of (a) bias solenoid and (b) gas valve at maximum discharge voltage.

A high degree of vacuum is the key to a successful discharge. Usually, to remove residual gas and water vapor from the interior, the electrodes are heated to 120°C by using heating tape and held for a period of time. The vacuum system consists of a mechanical pump and a molecular pump (FF-200), and the pumping speed is 1200 L/s. The base pressure is approximately 1×10^{-5} Pa.

2.2 Diagnostics

Table 2. Diagnostics of the CT injector system.

Diagnostics	Measurement parameter
Magnetic probe	Velocity/Magnetic field strength
Langmuir probe	Electron density/Temperature
Rogowski coil	Discharge current
Differential probe	Discharge voltage
CCD camera	Image
Impurity spectrometer	Impurity type

To measure the discharge waveforms and the relevant characteristics of the CT, several diagnostic systems were installed on the EAST-CTI system (table 2), including Langmuir probes, magnetic probes, a Rogowski coil, a differential probe, a charge-coupled device (CCD) camera, and an impurity spectrometer.

The locations of the diagnostic systems are shown in figure 1(a). To monitor the variation of CT density in the acceleration region, three Langmuir probes (L1–L3) are placed at z = 380, 570, and 790 mm. In the same positions in the axial direction, three magnetic probes (P1–P3) are used to monitor the poloidal magnetic field of the CT. The effective area of the magnetic probe is approximately 400 mm^2, and the frequency response can reach up to 1 MHz, thus, meeting the requirements for CT magnetic measurements. The magnetic probes include a passive integrator (τ_{RC} = 50µs), which allows the magnetic fluctuation signal to recover the magnetic field signal of the CT. The discharge current and voltage of the solenoid, gas valve, formation region, and acceleration region are monitored by Rogowski coils and a differential probe. To observe the plasma discharge process, a

CCD camera (Phantom V710 high-speed camera, resolution: 256×128, sampling rate: 150,000 fps, 1 µs open) is placed at the end of the injector for recording. To monitor the type of impurities in the CT, an impurity spectrometer (Wyoptics scientific class spectrometer MAX2000 pro) is placed in the side port of the injector. All signals from diagnostic systems are recorded on the acquisition system.

3 Experiment results

3.1 Typical discharge waveforms and formation thresholds of CT

The precise control of time is important for CT pulse discharge. For this purpose, a set of field-programmable gate array (FPGA)-based high-precision timing control modules was developed in this study. The injector system uses optical fibers for triggering, effectively protecting the system. The discharge process of the injector system is illustrated in figure 3(a). The control system gives charging instructions to the charger through the switch and receives the returned charging status. Then, the triggering of each power system is completed sequentially by the optical fiber in the order of discharge.

A typical discharge sequence is depicted in figure 3(b). The bias solenoid discharge time was set as −7000 µs, as shown in figure 3(b1). The bias current was approximately 40 A with the charging voltage V_{sol} as 0.8 kV, the poloidal magnetic field was approximately 0.17 T, and the bias magnetic flux was approximately 3 mWb. The injection time of neutral gas is also crucial for the formation of the CT. At room temperatures, the adiabatic expansion velocity of helium in the formation region was approximately 1.24 km/s [17], and the distance from the gas valve to the end of the formation region was 0.25 m. To ensure uniform diffusion of neutral gases while preventing them from diffusing outside the formation region, the gas valve discharge duration was selected as 200 µs, which is earlier than the discharge of the formation bank (figure 3(b2)). The gas valve current was approximately 3700 A with the charging voltage V_{gas} as 1 kV, delivering up to 10^{20} particles at a time into the injector system.

The current and voltage of the formation bank and acceleration bank are key factors in CT formation and acceleration. As shown in figure 3(b3) and (b4), the moment of formation region discharge was considered as the zero moment. The formation and acceleration discharge currents were 144 and 87 kA, with the discharge voltage as 6.5 and 5.5 kV, respectively. The circuit efficiency was defined as $\eta = \int I_{\mathrm{gun}} \cdot V_{\mathrm{gun}} dt / \left(1/2 \cdot CU^2\right)$, where I_{gun} is the discharge current and V_{gun} is the discharge voltage. The circuit efficiency of the injector in the formation region and acceleration region was 37.3% and 86.4%, respectively. A necessary condition for the formation of the CT is $\lambda_f > \lambda_c$; here, $\lambda_f = \mu_0 I_f / \psi_{\mathrm{bias}}$ [18], $\lambda_c = 1/r_{\mathrm{in}} \cdot \sqrt{2/\ln\left(r_{\mathrm{out}}/r_{\mathrm{in}}\right)}$, and I_f is the formation discharge current. The value of λ_f is 60 m^{-1} in this discharge, and the desired value of λ_c depends on the size of the injector (30.3 m^{-1} in this study). Thus, the parameter values satisfied the requirements for the formation and acceleration of a CT.

3.2 Performance of EAST-CTI

Figure 4(b), (d), and (f) shows the relationship between acceleration region current and CT density at different locations in the acceleration region. The maximum electron density at $Z = 38$, 57, and 76 cm was 3.91×10^{20}, 6.16×10^{20}, and 8.73×10^{20} m^{-3}, respectively. The velocity values $v_{1,2}$ and $v_{2,3}$, which represent the velocity between $Z = 38$–57 cm and $Z = 57$–76 cm, were 39.3 and 56.0 km/s, respectively. The velocity growth rate of CT between these two distances was approximately 42.5%.

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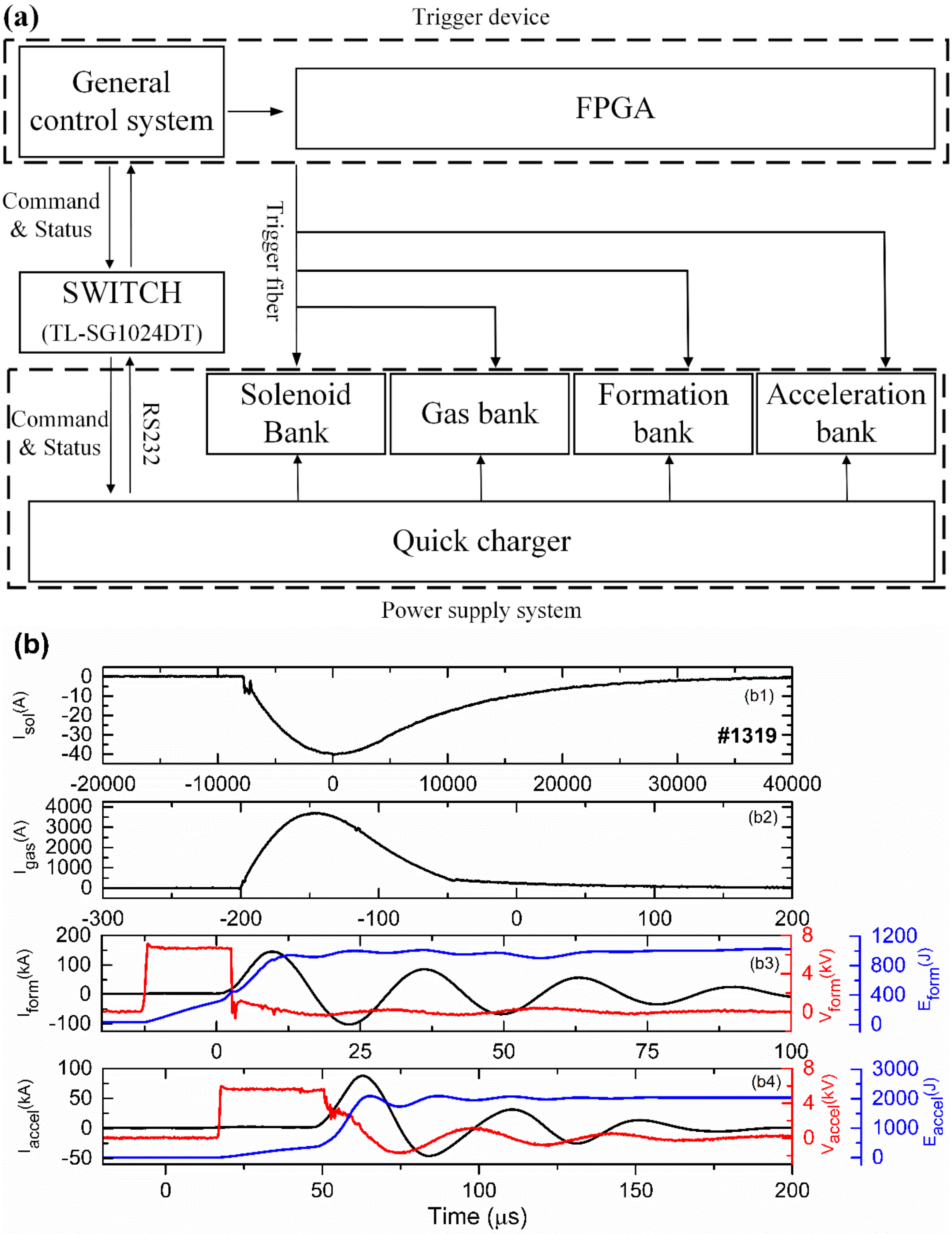

Figure 3. (a) Discharge process of the injector system, (b) typical discharge waveforms.

Moreover, the density of CT increased from 3.91×10^{20} m^{-3} at 38 cm to 8.73×10^{20} m^{-3} at 76 cm. The poloidal magnetic field B_p indicated that the CT had been formed. Three magnetic probes located in the acceleration region were used to measure the magnetic field. Figure 4(c), (e), (g) shows the changes in the boundary magnetic field at the same axial position. The negative peaks of B_{p1}, B_{p2}, and B_{p3} signals exhibited continuous decay, indicating that the CT decayed during transmission.

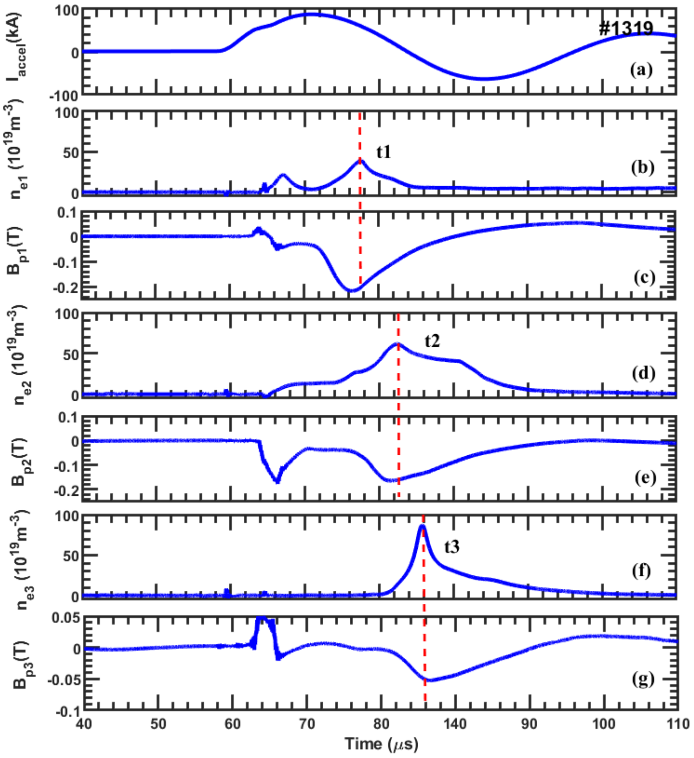

Figure 4. The experiment results of CT: (a) acceleration current, (b, d, f) electron density and (c, e, g) magnetic field strength.

To estimate the total number of particles, the equation $\int n_e dV = \int S_{\mathrm{CT}} v_{\mathrm{CT}} n_e dt$ was used, where V is the volume of CT, S_{CT} is the particle flow cross-sectional area ($\sim 0.007\,\mathrm{m}^2$). The total number of particles, mass, and length of the CT were $N_p \approx \int S_{\mathrm{CT}} V_{\mathrm{CT}} n_e(t) dt = 2.4 \times 10^{18}$, $m_{\mathrm{CT}} \approx 16.1\,\mu\mathrm{g}$, and $L \approx v_{\mathrm{CT}} \times t_{\mathrm{fwhm}} = 0.17\,\mathrm{m}$, respectively. The density of CT was assumed to be uniformly distributed. The injected momentum, total directional kinetic energy, and penetration pressure were $P_{\mathrm{CT}} \approx m_i N_p v_{ct} = 9 \times 10^{-4}\,\mathrm{N{\cdot}s}$, $E_k \approx \frac{1}{2} m_i N_p v_{\mathrm{CT}}^2 = 25\,\mathrm{J}$, and $P_k \approx \frac{1}{2} m_i n_e v_{\mathrm{CT}}^2 = 1.6 \times 10^3\,\mathrm{Pa}$, respectively, and the corresponding maximum penetrating magnetic field was $B_{\max} = \sqrt{2\mu_0 P_k} = 0.06\,\mathrm{T}$.

To study the formation and evolution of CT during the discharge, images were captured using a high-speed camera with an interval of 6.67 μs. The starting time of formation region discharge was selected as the zero-time trigger of the camera. As seen in figure 5(a)–(d), the breakdown of the neutral gas was observed at $t = 6.67\,\mu\mathrm{s}$, and four channels of current formed in the local area. Subsequently, the four plasma clusters spread rapidly in the radial direction and swept the neutral particles. At $t = 13.33\,\mu\mathrm{s}$, two bright clusters of light were observed in the images, rotating around the inner electrode. The bright cluster underwent a significant rotation at $t = 20\,\mu\mathrm{s}$ compared to $t = 13.33\,\mu\mathrm{s}$. The equation, $\mathrm{v}_\theta = 2\pi r_{\mathrm{in}}/\pi T \approx 37\,\mathrm{km/s}$, was used to determine the velocity of rotation, where θ is the angle of rotation and T is the interval between these times. The rapid rotation was possibly due to the bright plasma cluster subjected to a force $E_r \times B_z$ drift motion in the toroidal direction (formation electrode voltage = 1.2 kV, and bias field = 0.17 T). At 26.67 μs, no bright spots were noted on the plasma cluster, indicating that the isomer cluster was uniformly mixed.

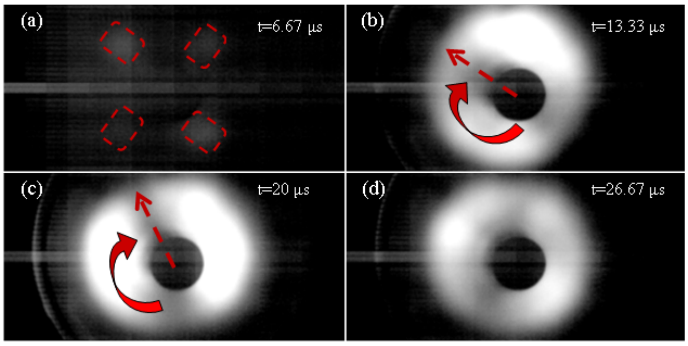

Figure 5. The images (a)–(d) of the CT rotating in the toroidally direction during the initial phase of discharge.

When the discharge voltage was increased further, more collisions occurred between the plasma and the device during the discharge, resulting in more types of impurities in the CT, thus affecting the quality and speed of the CT. The measurement of impurity radiation spectra during the discharge process was performed using an impurity spectrometer. The impurity radiation spectrum, that is, the impurity spectral lines of C, Fe, Cr, Cu, and Mg, were measured in the wavelength range of 900–1100 nm during the discharge (figure 6). Because the neutral gas was struck and ionization occurred mainly in the formation region, the formation region was mainly composed of stainless steel and PTFE, which led to a relatively strong impurity radiation spectrum of C^I (906.2 nm) and Fe^I (926.4 nm). Furthermore, the acceleration region was made of copper, but the plasma did not undergo strong collisions in this region, and the impurity spectrum of Cu^{II} (1027.9 nm) was weak. Impurities may be the main cause of CT magnetic field attenuation; impurities can also greatly affect the life of the CT. In future studies, we will coate chromium or tungsten for the inner wall of the injector to solve the problem of a large amount of metal impurities in the plasma.

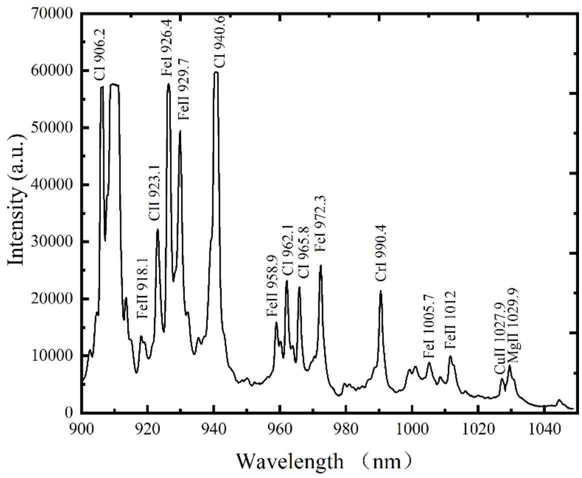

Figure 6. Measurement result of the dominant wavelength by using the impurity spectrometer.

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4 Summary

In this paper, we described a newly designed CT injector (i.e., EAST-CTI) for central fueling of EAST and performed a series of tests. In summary, the CT realized using newly developed EAST-CTI exhibits a velocity, electron density, particle inventory, mass, length, injected momentum, total directional kinetic energy, and penetration pressure of 56 km/s, 8.73×10^{20} m^{-3}, 2.4×10^{18}, 16.1 µg, 0.17 m, 9×10^{-4} N·s, 25 J, and 1.6×10^{3} Pa, respectively. In future studies, higher parameters will be obtained by increasing the discharge voltage and optimizing the experimental procedures.

Acknowledgments

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References

[1] Y.T. Song, S.T. Wu, J.G. Li, B.N. Wan, Y.X. Wan, P. Fu et al., *Concept design of CFETR tokamak machine*, *IEEE Trans. Plasma Sci.* **42** (2014) 503.

[2] L. Perkins, S. Ho and J. Hammer, *Deep penetration fuelling of reactor-grade tokamak plasmas with accelerated compact toroids*, *Nucl. Fusion* **28** (1988) 1365.

[3] P.B. Parks, *Refueling tokamaks by injection of compact toroids*, *Phys. Rev. Lett.* **61** (1988) 1364.

[4] X. Zheng, J. Li, J. Hu, J. Li, R. Ding, B. Cao et al., *Comparison between gas puffing and supersonic molecular beam injection in plasma density feedback experiments in EAST*, *Plasma Phys. Control. Fusion* **55** (2013) 115010.

[5] R. Raman, F. Martin, E. Haddad, M. St-Onge, G. Abel, C. Cote et al., *Experimental demonstration of tokamak fuelling by compact toroid injection*, *Nucl. Fusion* **37** (1997) 967.

[6] D. Liu, C. Xiao, A. Singh and A. Hirose, *Bench test and preliminary results of vertical compact torus injection experiments on the STOR-M tokamak*, *Nucl. Fusion* **46** (2005) 104.

[7] M. Nagata, H. Ogawa, S. Yatsu, N. Fukumoto, H. Kawashima, K. Tsuzuki et al., *Experimental studies of the dynamics of compact toroid injected into the JFT-2M tokamak*, *Nucl. Fusion* **45** (2005) 1056.

[8] T. Asai, T. Matsumoto, T. Roche, I. Allfrey, H. Gota, J. Sekiguchi et al., *Compact toroid injection fueling in a large field-reversed configuration*, *Nucl. Fusion* **57** (2017) 076018.

[9] C. Chen, T. Lan, C. Xiao, G. Zhuang, D. Kong, S. Zhang et al., *Development of a compact torus injection system for the Keda Torus eXperiment*, *Plasma Sci. Technol.* **24** (2022) 045102.

[10] M. Yamada, Y. Ono, A. Hayakawa, M. Katsurai and F.W. Perkins, *Magnetic reconnection of plasma toroids with cohelicity and counterhelicity*, *Phys. Rev. Lett.* **65** (1990) 721.

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[11] S. Howard, M. Laberge, L. McIlwraith, D. Richardson and J. Gregson, *Development of merged compact toroids for use as a magnetized target fusion plasma*, *J- Fusion Energy* **28** (2008) 156.

[12] Y. Zhang, D.M. Fisher, M. Gilmore, S.C. Hsu and A.G. Lynn, *Experimental investigation of coaxial-gun-formed plasmas injected into a background transverse magnetic field or plasma*, *Phys. Plasmas* **25** (2018) 055709.

[13] J. Li and Y. Wan, *The experimental advanced superconducting tokamak*, *Engineering* **7** (2021) 1523.

[14] J.B. Taylor, *Relaxation of toroidal plasma and generation of reverse magnetic fields*, *Phys. Rev. Lett.* **33** (1974) 1139.

[15] S.P. Chow, S. Lee and B.C. Tan, *Current sheath studies in a co-axial plasma focus gun*, *J. Plasma Phys.* **8** (1972) 21.

[16] T. Matsumoto, J. Sekiguchi, T. Asai, H. Gota, E. Garate, I. Allfrey et al., *Development of a magnetized coaxial plasma gun for compact toroid injection into the C-2 field-reversed configuration device*, *Rev. Sci. Instrum.* **87** (2016) 053512.

[17] D.C. Black, R.M. Mayo and R.W. Caress, *Direct magnetic field measurement of flow dynamics in magnetized coaxial accelerator channels*, *Phys. Plasmas* **4** (1997) 2820.

[18] C.W. Barnes, T.R. Jarboe, G.J. Marklin, S.O. Knox and I. Henins, *The impedance and energy efficiency of a coaxial magnetized plasma source used for spheromak formation and sustainment*, *Phys. Fluids B* **2** (1990) 1871.

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